17 The table shows some information about the dress sizes of 25 women.

Dress size	Number of women
8	2
10	9
12	8
14	6

(a) Find the median dress size.

Median position = (25 + 1) ÷ 2 = 13th.

Sizes 8 and 10 account for the first 11 women.

The 13th woman has dress size 12.

1 mark
Identify the middle observation

12 **(1)**

3 of the 25 women have a shoe size of 7

Zoe says that if you choose at random one of the 25 women, the probability that she has either a shoe size of 7 or a dress size of 14 is $\frac{9}{25}$ because

$$\frac{3}{25} + \frac{6}{25} = \frac{9}{25}$$

(b) Is Zoe correct?
You must give a reason for your answer.

No. A woman could have both shoe size 7 and dress size 14.

Adding the probabilities could count her twice.

(1)

1 mark
The events need not be mutually exclusive

(Total for Question 17 is 2 marks)

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